

KitchenStock — System Architecture

Multi-tenant Restaurant Inventory & Operations SaaS · Node.js · Express · MongoDB · React / React Native

530+ tests · 33 suites

CLIENT APPS

React Web App
owner & staff dashboard · all modules

React Native Mobile
counts · approvals · push

↓ HTTPS / JWT Bearer

EDGE · SECURITY · MIDDLEWARE

CORS

Helmet (CSP)

Rate Limiting

Mongo-Sanitize

HPP

JWT auth + tokenVersion

RBAC / Feature flags

AI data-egress opt-in

EXPRESS API — ROUTE DOMAINS (/api/*, 30+ routers)

Auth & Identity
OTP · JWT+tv · sessions
role-based access control

Inventory
items · adjustments · audit
transfers · batches / expiry

Purchasing
suppliers · POs · receiving
invoices + AI/OCR extract

Recipes & Production
costing · prep sheets
available-to-produce calc

Sales & POS
sales log · deductions
POS CSV import + mapping

Analytics & Reports
valuation · dead-stock
profitability · price trends

Forecasting
demand · depletion
reorder → auto-draft PO

AI Copilot · RAG
grounded Q&A · insights
OpenAI / Anthropic (pluggable)

Billing & Plans
Stripe checkout + portal
webhooks · signed events

Notifications
push · email alerts
low-stock / expiry scan

DOMAIN LOGIC · SHARED LIBS

recipeExpand (BOM)

forecast (moving-avg)

aiContext · RAG retrieval

deductRecipe

observability (Sentry)

csv · pdf/OCR · units convert

DATA LAYER

MongoDB · Mongoose
25+ models · every collection scoped by organizationId (multi-tenant isolation)

Users/RBACItems/StockSuppliers/POInvoicesRecipesWasteBatchesHistory/Audit

External Integrations

StripeTwilio SMS

OpenAISMTTP Email

AnthropicExpo Push

SentryPDF/OCR

ENGINEERING & SKILLS DEMONSTRATED

Node.js

Express

MongoDB / Mongoose

REST API design

Multi-tenant SaaS

JWT / Auth / RBAC

Stripe Billing

AI / RAG / LLM

Jest · 530+ tests

Security hardening

React / RN

Forecasting / OCR

Observability / Ops

Domain modeling (BOM/units)